



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.NOS.9

Represent Polygons on the Coordinate Plane

Lesson 7-7 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

Language Objective: I can explain how I found a distance using the words vertices, coordinates, and absolute value.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

vertex

polygon

perimeter

distance between vertices

scale



Tap any word to see what it means and a picture.

1

— A corner point of a polygon, named by an ordered pair on the coordinate plane.

2

— A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.

3

— The total distance around a polygon — the sum of all its side lengths.

4

— When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.

5

— What one grid unit stands for in the real situation.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Where do the other two vertices go — and how do the known coordinates, the scale, and the area work together to tell you?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **vertex** — A corner point of a polygon, named by an ordered pair on the coordinate plane.
vértice — Un punto de esquina de un polígono, nombrado por un par ordenado en el plano de coordenadas.

☐ **polygon** — A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.
polígono — Una figura cerrada hecha de segmentos; en el plano se dibuja marcando los vértices y conectándolos en orden.

☐ **perimeter** — The total distance around a polygon — the sum of all its side lengths.
perímetro — La distancia total alrededor de un polígono: la suma de todos sus lados.

☐ **distance between vertices** — When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.
distancia entre vértices — Cuando dos vértices comparten una coordenada x o y, la distancia es la diferencia absoluta de las otras coordenadas.

☐ **scale** — What one grid unit stands for in the real situation.
escala — Lo que representa una unidad de la cuadrícula en la situación real.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The park's width is ____ feet. Using the area formula, its length is ____ feet, which is ____ units on the map — so the new vertices sit that many units below the known ones.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: When two vertices share the same y-coordinate, the segment between them is horizontal, so its length comes from the DIFFERENCE in the coordinate that changes — the x-coordinates.

Evidence: A strong answer explains that the SHARED coordinate tells you the direction of the side, and the DIFFERING coordinate is what you subtract. A weak answer just says 'subtract the numbers' without identifying which pair.

Reasoning: This evidence connects the definition of polygon to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The park's width is ____ feet. Using the area formula, its length is ____ feet, which is ____ units on the map — so the new vertices sit that many units below the known ones.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- I know because ____.
Lo sé porque ____.
- So ____.
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
 - **My math evidence is ____.**
Mi evidencia matemática es ____.
 - **This proves ____ because ____.**
Esto demuestra ____ porque ____.
-
-
-

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** vertex
2. **Notes 2:** polygon
3. **Notes 3:** perimeter
4. **Notes 4:** distance between vertices
5. **Notes 5:** scale
6. **Watch for this mistake:** *Subtracting coordinates without the absolute value — from $y = 3$ down to $y = -3.75$ the distance is $3 + 3.75 = 6.75$ units, not $3.75 - 3 = 0.75$. Distances that cross zero are the whole reason the absolute value is there.*
7. **Try It 1:**
8. **Try It 2:**